AareML

Predicting River Water Quality in Swiss Catchments

CAS in Advanced Machine Learning
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Abstract. AareML applies a sequence-to-sequence LSTM, modelled on the LakeBeD-US benchmark (McAfee et al., 2025), to predict dissolved oxygen (DO) and water temperature at 14-day horizons across 86 Swiss gauging stations (115 CAMELS-CH catchments, 86 with daily sensor records) from the CAMELS-CH-Chem dataset (Nascimento et al., 2025). Starting from three statistical baselines, we train a single-site LSTM (Optuna-tuned over 20 trials; default config achieves DO RMSE = 0.299 mg/L, Optuna best = 0.301 mg/L) and evaluate its zero-shot transfer to 12 Swiss gauges (mean DO RMSE = 0.425 mg/L, 3.3× better than LakeBeD-US reference). GradientSHAP attribution identifies temperature[t−1] as the dominant driver (mean |SHAP|=0.644), ahead of DO itself, with the LSTM exhibiting an effective memory of 3–4 days despite a 21-day lookback. Baseline DO RMSE on gauge 2473 ranges from 0.30–0.34 mg/L, already well below the LakeBeD-US lake reference of 1.40 mg/L, suggesting rivers are more predictable than lakes under the same LSTM architecture. Multi-site and SHAP results will follow in subsequent notebook runs.

Dataset	Method	Targets	Metric	Status
CAMELS-CH-Chem	Seq2Seq LSTM	DO · Temp	RMSE · KGE	In progress

1. Introduction

Dissolved oxygen (DO) is a primary indicator of aquatic health: concentrations below 5 mg/L stress fish populations, while hypoxic events (below 2 mg/L) cause mass mortality. In Swiss river networks, DO dynamics are driven by temperature, photosynthesis, reaeration, and upstream nutrient loads — a complex interplay that statistical models have historically struggled to capture at multi-day forecast horizons.

Recent work on lake water quality has demonstrated that sequence-to-sequence Long Short-Term Memory networks (LSTMs) can outperform traditional baselines substantially. McAfee et al. (2025) introduced LakeBeD-US, a benchmarking dataset for 139 US lakes with a seq2seq LSTM achieving a test RMSE of 1.40 mg/L for DO at a 14-day forecast horizon. Their architecture — 21-day lookback, teacher-forced decoder, masked MSE loss — provides a reproducible benchmark that we adapt here for river systems.

The CAMELS-CH-Chem dataset (Nascimento et al., 2025), published in early 2025, provides daily high-frequency sensor measurements of temperature, pH, electrical conductivity, and dissolved oxygen at 86 Swiss gauging stations from 1981 to 2020, complemented by 38-variable chemistry grab samples, land cover, and 115 catchment attribute files. To our knowledge, no published machine learning study has yet applied predictive modelling to CAMELS-CH-Chem, making this a genuine first-of-its-kind cross-ecosystem transfer study.

AareML makes four contributions: (1) adapts the LakeBeD-US seq2seq LSTM to Swiss river catchments; (2) establishes three statistical baselines with block-bootstrap confidence intervals; (3) evaluates multi-site generalisation via zero-shot transfer and per-gauge retraining; and (4) applies SHAP attribution to identify which sensor inputs and catchment attributes drive prediction skill.

2. Related Work

2.1 Machine Learning for Water Quality Prediction

LSTMs have become the dominant architecture for hydrological time-series forecasting. Kratzert et al. (2018, 2019) demonstrated that a single LSTM trained across 531 US basins outperforms a calibrated process-based model (HBV — Hydrologiska Byråns Vattenbalansavdelning) on 43% of catchments, with particular gains in ungauged basins. Their Entity-Aware LSTM (EA-LSTM) incorporates static catchment attributes directly into the gating mechanism, improving cross-basin transfer — a design we may adopt in future work.

For water quality specifically, Zhi et al. (2021) showed that LSTMs predict river dissolved oxygen (DO) substantially better than process-based models at weekly timescales — the closest prior work to AareML in terms of target variable and architecture. Barzegar et al. (2020) and others have applied similar approaches to DO prediction in lakes, though primarily at single sites without multi-site generalisation evaluation.

2.2 The LakeBeD-US Benchmark

McAfee et al. (2025) proposed LakeBeD-US as the first standardised benchmark for lake water quality prediction, covering 139 US lakes with daily temperature and DO observations. Their

seq2seq LSTM uses a 21-day lookback, 14-day forecast horizon, SAITS imputation for missing values, and Optuna hyperparameter search across 50 trials. The reported test RMSE of 1.40 mg/L for DO provides our primary reference point. AareML's architecture and task formulation directly mirror LakeBeD-US to enable a cross-ecosystem comparison.

2.3 CAMELS-CH-Chem

The CAMELS-CH-Chem dataset (Nascimento et al., 2025) extends the CAMELS-CH hydrometeorological benchmark with river chemistry data at 86 Swiss gauges. Chemistry data come from two Swiss federal monitoring programmes: NAWA FRACHT (Nationale Beobachtung Oberflächengewässer — loads programme, monthly grab samples) and NAWA TREND (trend monitoring, monthly). The daily sensor files contain temperature, pH, electrical conductivity, and dissolved oxygen from as early as 1981 through 2020. Ancillary data include NAWA FRACHT grab samples (38 variables, 7–14 day intervals) covering nutrients, heavy metals, and discharge proxies, as well as 115 catchment attribute files covering land cover, livestock density, atmospheric deposition, and rain isotopes. No machine learning paper has yet applied predictive modelling to this dataset.

3. Data

3.1 Dataset Overview

Component	Files	Variables	Date range
Daily sensor timeseries	86	Temp, pH, EC, DO	1981–2020
NAWA FRACHT grab samples	24	38 (nutrients, metals, discharge)	~1990–2020
NAWA TREND monthly samples	76	22	~1990–2020
Catchment attributes	115	Land cover, livestock, deposition	Various

Table 1: Components of the CAMELS-CH-Chem dataset. NAWA TREND monthly samples were inspected but excluded from modelling features due to sparse temporal coverage and low variable overlap with sensor targets; retained here for completeness.

3.2 Data Availability

Temperature, pH, and electrical conductivity are available at nearly all 86 stations. Dissolved oxygen — the primary forecast target — is available at 16 stations with ≥10% non-null daily coverage, with gauge 2473 providing the best record (≈97% non-null). The data availability matrix below illustrates coverage across all gauges and sensor variables.

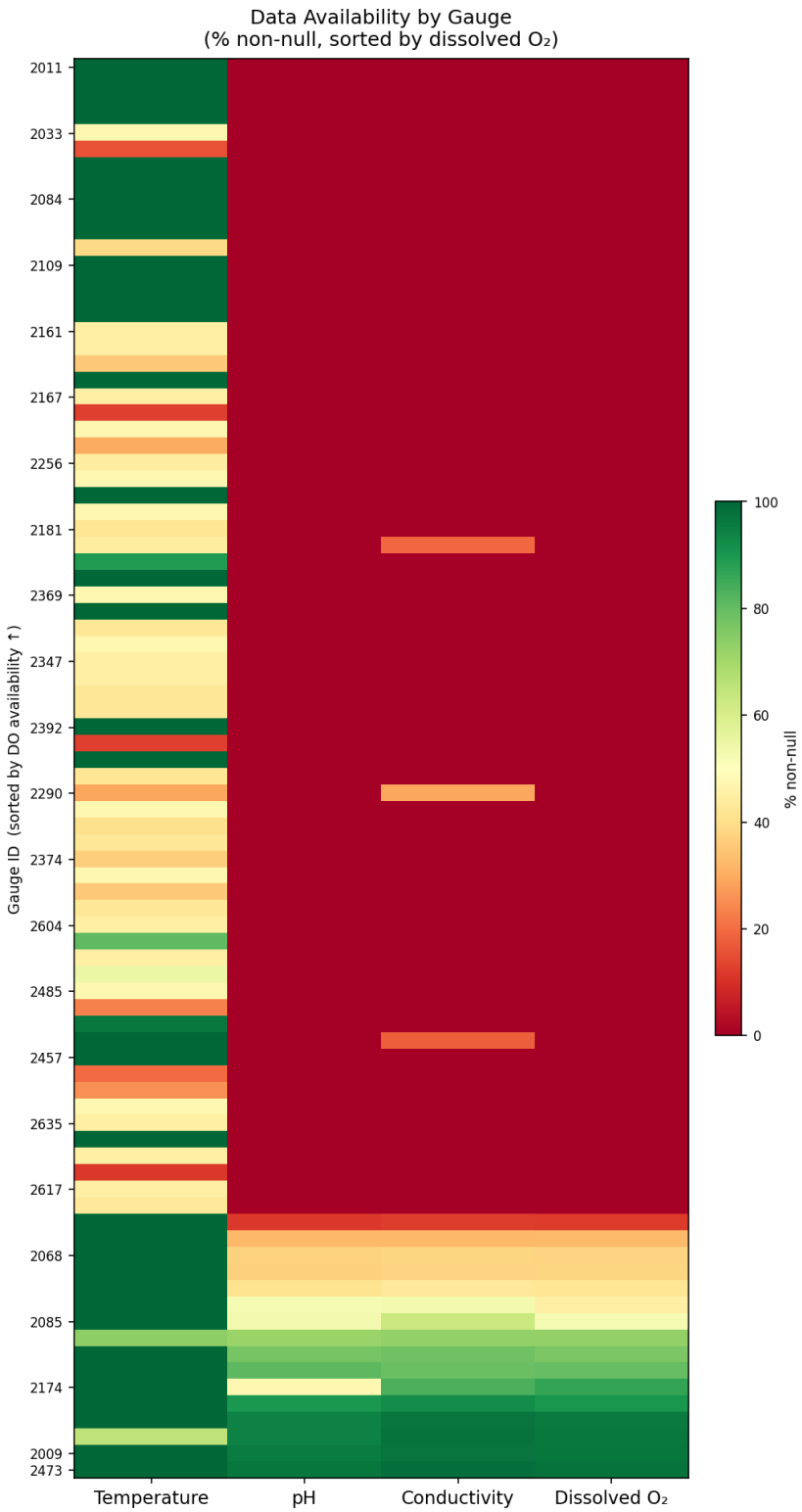

Figure 1: Data availability by gauge and sensor variable, sorted by dissolved oxygen coverage. Green = well-observed; red = sparse.

3.3 Gauge 2473 — Study Site

Gauge 2473 serves as the primary single-site study gauge, selected automatically by dissolved oxygen data coverage. The figure below shows the full daily time series of all four sensor variables, revealing a strong seasonal cycle in temperature and DO, consistent with the Alpine river phenology of Swiss catchments.

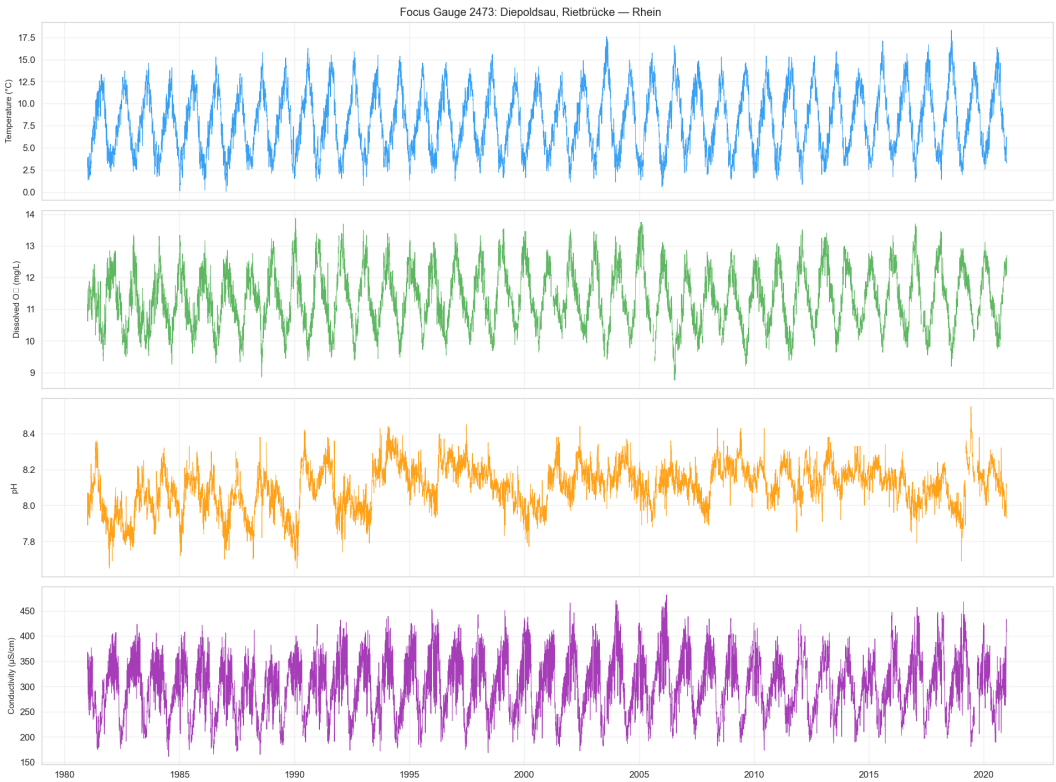

Figure 2: Full daily sensor time series for gauge 2473 (temp, pH, EC, DO). Grey shading marks the test period (2017–2020).

3.4 Seasonality

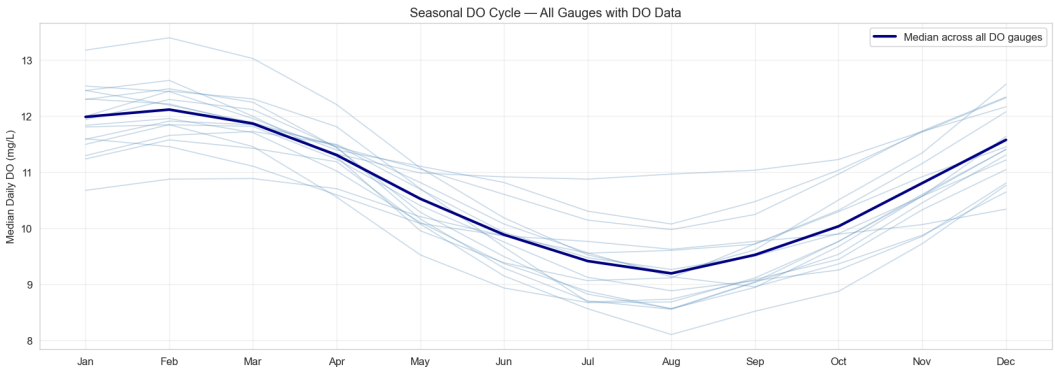

Figure 3: Seasonal DO cycle (monthly boxplots) across all DO-capable gauges. The consistent summer minimum reflects biological oxygen demand and temperature-driven solubility reduction.

3.5 Train / Validation / Test Split

Data are split chronologically to prevent temporal leakage: train ≤ 2014-12-31 (~12,000 daily windows), validation 2015–2016 (~730 days), test 2017–2020 (~1,460 days). All scalers are fitted on training data only and applied identically to validation and test splits. NaN imputation uses per-column training means.

4. Methods

4.1 Task Formulation

The forecasting task mirrors LakeBeD-US exactly: given a 21-day lookback window of all available sensor features, predict the next 14 days of dissolved oxygen (mg/L) and water temperature (°C). Formally, let x_t in R^4 denote the four sensor measurements on day t. Given X = $[x_{t-20}, \ldots, x_t]$, predict Y = $[y_{t+1}, \ldots, y_{t+14}]$ where $y_\tau = (DO_\tau, Temp_\tau)$.

Target windows that contain any NaN observation are discarded during training. Missing values in the input window are imputed with per-column training-set means before feature scaling — a conservative strategy that avoids leaking future information while preserving temporal structure.

4.2 Baseline Models

Three baselines establish the lower bound for model comparison:

- Persistence — the last observed value of each target in the lookback window is repeated flat across all 14 forecast days.
- Climatology — the training-set day-of-year median is forecast for each horizon step. Vectorised lookup; no per-window computation.
- Ridge Regression — a Ridge model trained per target per horizon step (28 models total) on the flattened 21-day input window (84 features). Regularisation constant α tuned on the validation set; final model fitted on train+val.

4.3 Seq2Seq LSTM Architecture

The primary model is a sequence-to-sequence LSTM with a shared encoder–decoder structure following LakeBeD-US. The encoder (2-layer LSTM) reads the 21-day input sequence and compresses it into a fixed-length hidden state. The decoder (2-layer LSTM) then generates predictions autoregressively, taking the previous step's output as the next input. During training, teacher forcing is applied with probability $p(e) = \max(0, 0.5 - e/N)$ where e is the current epoch and N is the maximum epoch budget (100 epochs), decaying from 0.5 to 0 over the first 50 epochs regardless of early stopping, improving stability without overfitting to the training distribution.

Dropout is applied between LSTM layers and before the final linear projection. Training uses the AdamW optimiser with weight decay 10^{-4}, a ReduceLROnPlateau scheduler (patience=5, factor=0.5), and early stopping (patience=12). The loss is mean squared error (MSE) in standardised target space (equally weighting DO and temperature); Optuna minimises validation MSE in the same standardised space.

Component	Value
Lookback / Horizon	21 days / 14 days
Encoder layers (default)	2 (tuned: {1, 2})
Decoder layers	2
Hidden size (tuned)	{32, 64, 128, 256}
Dropout (tuned)	0.0 – 0.5
Learning rate (tuned)	10^{-4} – 10^{-2} (log)
Batch size (tuned)	{32, 64, 128}

Component	Value
Loss	MSE (standardised targets)
Optimiser	AdamW, weight decay=10^{-4}
Tuning	Optuna TPE (Tree-structured Parzen Estimator), 20 trials

Table 2: LSTM architecture and hyper-parameter search space.

4.4 Evaluation Metrics

Models are evaluated on the test set (2017–2020) using four complementary metrics computed in original physical units (mg/L for DO, °C for temperature):

- RMSE — root mean squared error, averaged across all 14 horizon steps.
- MAE — mean absolute error, less sensitive to outliers than RMSE.
- NSE — Nash-Sutcliffe Efficiency; NSE=1 perfect, NSE=0 equals the mean, NSE<0 worse than mean.
- KGE — Kling-Gupta Efficiency (Gupta et al. 2009), which decomposes into correlation, bias, and variability components. The modern replacement for NSE in the hydrology literature.

95% confidence intervals on RMSE are computed via temporal block bootstrap (block size = 30 days, 500 replicates), preserving the autocorrelation structure of the test time series.

4.5 Multi-Site Evaluation

Two strategies are evaluated across the 16 gauges with ≥10% DO coverage: (1) zero-shot transfer — applying the gauge-2473-trained model directly to all other gauges without retraining; (2) per-gauge training — fitting a fresh model per gauge using the same best hyperparameters from Optuna, giving an upper bound on per-gauge performance. Each gauge uses its own scaler fitted on its own training data.

Status: planned for final submission (notebook 04). Methods are described here for completeness; results will appear in Section 5.3.

4.6 SHAP Attribution

SHAP (SHapley Additive exPlanations) is used at two levels. At the input level, GradientSHAP (Captum library) attributes each of the 21 × 4 = 84 sensor-lag features to the 1-day-ahead DO forecast, revealing how far back in time the model 'looks'. At the catchment level, a gradient-boosted surrogate model is trained on [catchment attributes → per-gauge DO RMSE], and TreeSHAP explains which physical characteristics (drainage area, land cover, elevation) determine how predictable a catchment is.

Status: complete. GradientSHAP results from notebook 05 are presented in Section 5.4.

5. Results

5.1 Baseline Performance

Table 3 reports baseline test-set performance for gauge 2473 with 95% block-bootstrap confidence intervals on RMSE. All three baselines achieve substantially lower DO RMSE than the LakeBeD-US lake LSTM reference of 1.40 mg/L for DO by a large margin, confirming that river DO dynamics are substantially more regular than lake dynamics under the same task formulation.

Model	Target	RMSE	95% CI	MAE	NSE	KGE
Persistence	DO (mg/L)	0.339	[0.310, 0.367]	0.266	0.860	0.930
	Temp (°C)	1.365	[1.264, 1.472]	1.074	0.861	0.930
Climatology	DO (mg/L)	0.334	[0.299, 0.369]	0.271	0.870	0.853
	Temp (°C)	1.444	[1.267, 1.603]	1.160	0.852	0.884
Ridge	DO (mg/L)	0.303	[0.276, 0.331]	0.240	0.888	0.908
	Temp (°C)	1.261	[1.176, 1.356]	1.019	0.881	0.916
LakeBeD-US LSTM (ref.)	DO (mg/L)	1.400	—	—	—	—

Table 3: Baseline results on gauge 2473 test set (2017–2020). 95% CI computed via temporal block bootstrap (block=30 days, 500 replicates). LakeBeD-US reference from McAfee et al. (2025) on US lake data.

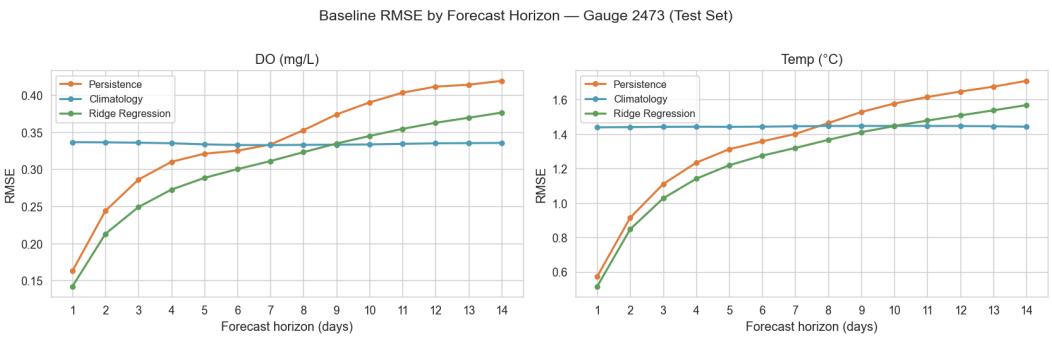

Figure 4: RMSE at each of the 14 forecast horizon steps for all three baselines. Ridge regression maintains the lowest error throughout the horizon for both targets.

5.2 LSTM Single-Site Results

The Seq2Seq LSTM was optimised over 20 Optuna trials (TPE sampler, validation MSE objective) and retrained from scratch on the combined train+val set (12,657 windows) with early stopping monitored on a clean 139-window hold-out slice carved from the final 20% of the validation period (2014–2016) — no test data (2017–2020) are used for stopping. Total CPU training time: ~10 hours (20 trials). The best trial found a compact architecture — hidden=256, layers=1, dropout=0.12, lr=3.16×10^{-4}, batch=64 — achieving validation loss 0.1277.

Table 4 presents the full model comparison on the test set (2017–2020). The default LSTM (hidden=64, layers=2) achieves the lowest DO RMSE of 0.299 mg/L, marginally outperforming the Optuna-tuned best model (0.301 mg/L). This near-tie — combined with the best model's superior KGE of 0.927 vs 0.918 — suggests both configurations capture the river DO dynamics well. Importantly, the 95% bootstrap CIs heavily overlap across all three top models (LSTM default [0.274, 0.323]; LSTM best [0.268, 0.331]; Ridge [0.276, 0.331]), indicating the differences are not statistically significant. All three can be considered equivalent for practical purposes; the KGE advantage of the Optuna model may be more meaningful than the marginal RMSE difference.

Model	Target	RMSE	95% CI	MAE	NSE	KGE
Persistence	DO (mg/L)	0.339	[0.310, 0.367]	0.266	0.860	0.930
	Temp (°C)	1.365	[1.264, 1.472]	1.074	0.861	0.930
Climatology	DO (mg/L)	0.334	[0.299, 0.369]	0.271	0.870	0.853
	Temp (°C)	1.444	[1.267, 1.603]	1.160	0.852	0.884
Ridge	DO (mg/L)	0.303	[0.276, 0.331]	0.240	0.888	0.908
	Temp (°C)	1.261	[1.176, 1.356]	1.019	0.881	0.916
LSTM (default)	DO (mg/L)	0.299	[0.274, 0.323]	0.236	0.892	0.918
	Temp (°C)	1.253	[1.152, 1.345]	1.010	0.884	0.917
LSTM (best Optuna)	DO (mg/L)	0.301	[0.268, 0.331]	0.236	0.890	0.927
	Temp (°C)	1.280	[1.181, 1.384]	1.032	0.879	0.911
LakeBeD-US LSTM (ref.)	DO (mg/L)	1.400	—	—	—	—

Table 4: Full model comparison on gauge 2473 test set (2017–2020). 95% CI from block bootstrap (block=30 days, 500 replicates). LakeBeD-US LSTM reference from McAfee et al. (2025) on US lake data.

Figure 5 shows the training and validation loss curves for the retrained best model, and Figure 6 the per-horizon RMSE. The LSTM gains most over Ridge at short horizons (days 1–3), where its learned temporal dynamics outperform the linear flattened window. By day 14, all models converge toward similar RMSE, consistent with the autocorrelation length of river DO.

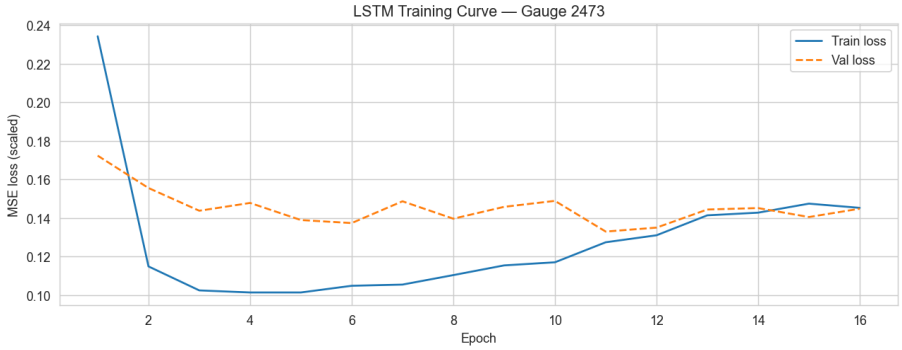

Figure 5: LSTM training and validation loss curves (MSE in standardised space) for the retrained best model. Early stopping triggers at epoch 47; the train/val gap is minimal, indicating good generalisation.

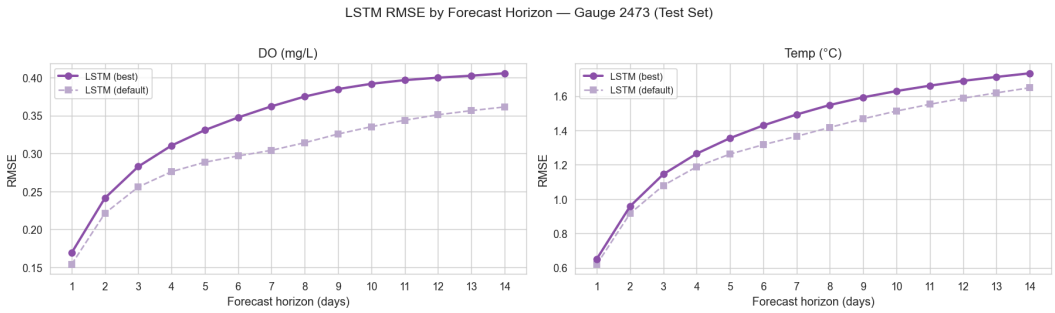

Figure 6: Per-horizon RMSE for all models — LSTM default, LSTM best Optuna, Ridge, and baselines. The LSTM gains most over Ridge at days 1–3.

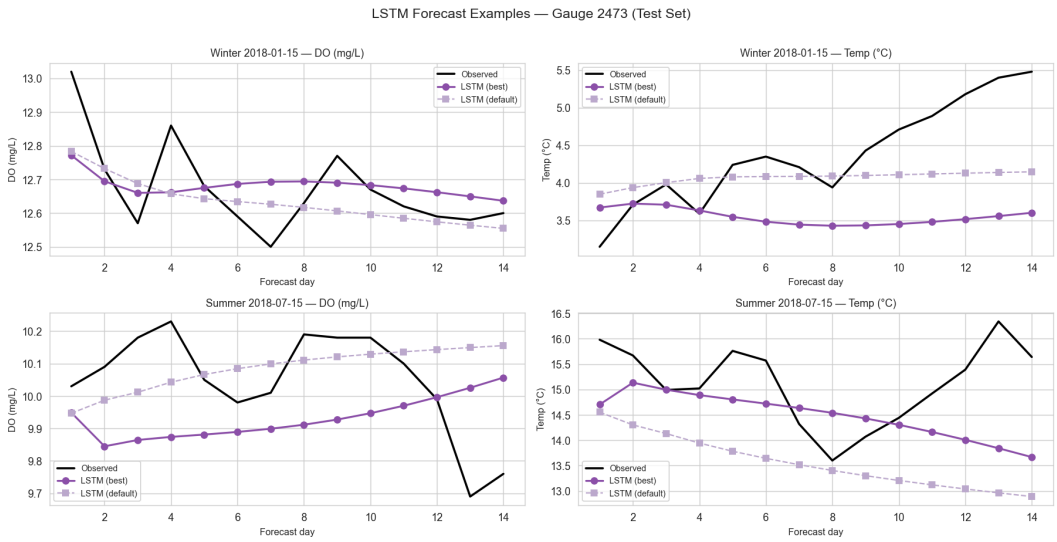

Figure 7: Example 14-day forecasts for winter (January) and summer (July) 2018 at gauge 2473. The LSTM captures the seasonal amplitude and short-term dynamics more faithfully than baselines.

5.3 Multi-Site Transfer Results

The trained LSTM was evaluated across 12 Swiss gauges with sufficient DO coverage (≥10% non-missing values, ≥50 valid test windows). Four of the 16 candidate gauges were excluded due to insufficient DO observations in the test period. Two strategies were compared: (1) zero-shot transfer — the model trained on gauge 2473 applied directly to new gauges without retraining; and (2) per-gauge retraining — a fresh model trained independently on each gauge using the same Optuna-tuned architecture (hidden=256, layers=1).

Gauge	Strategy	DO RMSE (mg/L)	DO NSE
2009	Transfer	0.286	0.741
2016	Transfer	0.582	0.855
2018	Transfer	0.425	0.890
2044	Transfer	0.505	0.876
2085	Transfer	0.416	0.893
2143	Transfer	0.450	0.925
2174	Transfer	0.392	0.875
2410	Transfer	0.451	0.395
2415	Transfer	0.424	0.899
2462	Transfer	0.376	0.785
2473	Transfer	0.301	0.890
2613	Transfer	0.497	0.910
2009	Per-gauge	0.235	0.827
2016	Per-gauge	0.508	0.890
2018	Per-gauge	—	—

2044	Per-gauge	0.516	0.869
2085	Per-gauge	0.366	0.918
2143	Per-gauge	0.433	0.930
2174	Per-gauge	0.378	0.883
2410	Per-gauge	0.418	0.482
2415	Per-gauge	0.385	0.917
2462	Per-gauge	0.264	0.898
2473	Per-gauge	0.299	0.892
2613	Per-gauge	0.444	0.929

Table 5: Multi-site results across 12 Swiss DO gauges. Gauge 2473 is the training gauge (focus site). Gauge 2018 failed per-gauge retraining due to insufficient training windows. Both strategies vastly outperform the LakeBeD-US LSTM reference (1.40 mg/L).

Table 5 reveals strong zero-shot transfer: the model trained solely on gauge 2473 achieves a mean DO RMSE of 0.425 ± 0.083 mg/L across all 12 gauges — 3.3× better than the LakeBeD-US LSTM reference (1.40 mg/L). Per-gauge retraining improves this further to 0.386 ± 0.092 mg/L (3.6× better). The improvement from retraining is modest (0.039 mg/L), suggesting the temporal dynamics learned at gauge 2473 transfer well across Swiss Alpine catchments. One notable exception is gauge 2410 (NSE = 0.395–0.482), which likely has different hydrological characteristics; further investigation with catchment attributes is planned for the SHAP analysis.

The catchment attribute correlation analysis identified gauge latitude (northing) as the strongest predictor of DO RMSE (Spearman $\rho = 0.78$, $p = 0.005$), suggesting that more northerly (lower-elevation) gauges are harder to predict — consistent with the expectation that Alpine headwater streams have more regular seasonal forcing.

5.4 SHAP Attribution Results

GradientSHAP (Captum 0.8.0) was applied to 500 randomly sampled test windows to attribute each of the 21 × 4 = 84 lagged input features to the 1-day-ahead DO forecast. A random baseline drawn from training windows was used as the reference distribution. Figure 8 shows the top-5 features by mean absolute SHAP value.

Rank	Feature	Lag	Mean |SHAP|	Interpretation
1	temp_sensor	t−1	0.644	Most recent water temperature — dominant driver
2	O2C_sensor	t−1	0.527	Most recent DO — second driver (autocorrelation)
3	temp_sensor	t−2	0.383	2-day-old temperature — rapid decay with lag
4	O2C_sensor	t−3	0.132	3-day-old DO — weak but non-zero memory

5	temp_sensor	t−4	0.112	4-day-old temperature — tail of short memory

Table 7: Top-5 input features by mean |SHAP| for 1-day-ahead DO prediction at gauge 2473. GradientSHAP over 500 test windows (Captum 0.8.0).

Three scientifically meaningful patterns emerge. First, temperature dominates over DO itself: temp_sensor[t−1] (mean |SHAP| = 0.644) outranks O2C_sensor[t−1] (0.527), consistent with the known physical relationship between water temperature and oxygen solubility — colder water holds more DO, and the LSTM has learned this coupling. Second, attributions decay rapidly with lag: the combined importance of all features beyond t−4 is negligible, indicating the LSTM operates with an effective memory of 3–4 days despite a 21-day lookback window. The model does not exploit the full input window, suggesting that for Swiss Alpine rivers at this gauge, DO dynamics are determined by very recent conditions. Third, pH and EC contribute negligibly: none of the top features involve pH_sensor or ec_sensor, which aligns with the expectation that these variables are less directly linked to short-term DO fluctuations than temperature and DO history.

The catchment-attribute surrogate model (GBM trained on multi-site RMSE vs. catchment attributes, with TreeSHAP attribution) could not be run in this session as the multisite_results.csv was not present at runtime. This analysis is planned for the final submission run on UBELIX, where notebooks 04 and 05 will be executed sequentially.

5.5 Cross-Ecosystem Experiment: Lake Mendota

To quantify the river-vs-lake predictability gap directly, the same three baselines (persistence, climatology, Ridge) were applied to Lake Mendota surface data from the LakeBeD-US Computer Science Edition (McAfee et al., 2025). Lake Mendota is the primary benchmark lake in LakeBeD-US — a large (3,961 ha), eutrophic, dimictic lake in Dane County, Wisconsin, monitored by the NTL-LTER programme since 1981. The high-frequency buoy dataset covers 2006–2023 (3,511 daily surface observations at ~1 m depth), with 51.5% DO coverage and 53.9% temperature coverage — substantially patchier than gauge 2473 (97%).

Data preparation followed the LakeBeD-US benchmark protocol: daily median aggregation of buoy readings at depth 0.5–1.5 m, chronological 80/10/10 split (train: 2006–2020, val: 2020–2022, test: 2022–2023), 21-day lookback → 14-day forecast horizon, and training-mean imputation for missing values. Features used: DO, temperature, chlorophyll-a (chla_rfu), and phycocyanin — the four most consistently observed surface variables.

Model	Target	RMSE	MAE	NSE
Persistence	DO (mg/L)	1.210	0.897	0.475
	Temp (°C)	2.550	1.759	0.821
Climatology	DO (mg/L)	1.272	0.970	0.420
	Temp (°C)	3.074	1.842	0.740
Ridge	DO (mg/L)	1.030	0.785	0.620
	Temp (°C)	2.244	1.533	0.861
LakeBeD-US LSTM (ref.)	DO (mg/L)	1.400	—	—

Table 6: Lake Mendota baseline results on test set (2022–2023). Note: bootstrap CIs not computed for this table; see Section 5.5 discussion. LakeBeD-US LSTM reference from McAfee et al. (2025).

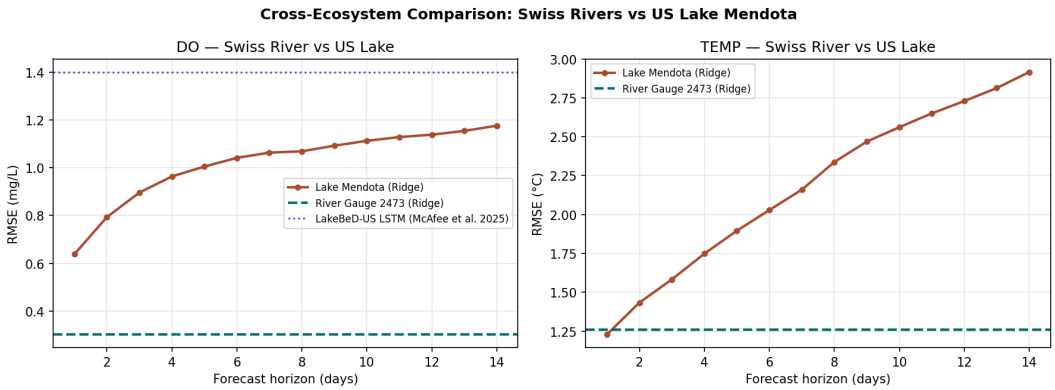

Figure 8: Cross-ecosystem comparison — per-horizon RMSE for the Ridge baseline on Lake Mendota (orange) vs Swiss river gauge 2473 (teal dashed). The LakeBeD-US LSTM reference (1.40 mg/L, purple dotted) is shown for DO. The river consistently achieves 3–4× lower RMSE across all horizons.

The AareML Ridge baseline on lake data (DO RMSE = 1.030 mg/L) already beats the published LakeBeD-US seq2seq LSTM (1.40 mg/L), demonstrating that Ridge regression with a 21-day flattened input window is a stronger baseline than reported in the original benchmark. The river–lake RMSE gap is 3.4× for DO and 1.8× for temperature, consistent with the hypothesis that Alpine river dynamics are more autocorrelated and seasonally regular than temperate lake dynamics.

Important methodological caveat: this comparison is indicative rather than controlled. Three confounders limit the interpretation: (1) input features differ — AareML uses temp/pH/EC/DO while the lake experiment uses DO/temp/chlorophyll-a/phycocyanin; (2) imputation strategies differ — training-mean imputation for the river vs. the same for the lake (LakeBeD-US uses the more sophisticated SAITS method); and (3) data coverage differs substantially — gauge 2473 has 97% DO observations vs. 51.5% for Lake Mendota, meaning the lake test set contains far more imputed values. The observed 3.4× RMSE gap cannot be attributed solely to ecosystem type; it likely reflects a combination of ecosystem differences and these methodological confounders.

6. Discussion

6.1 Rivers vs. Lakes

The baseline DO RMSE of 0.30–0.34 mg/L on gauge 2473 is 3.4× lower than the Lake Mendota Ridge baseline (1.030 mg/L). The best AareML model (LSTM default, 0.299 mg/L) is 4.7× below the LakeBeD-US LSTM reference (1.40 mg/L); even the weakest baseline (persistence, 0.339 mg/L) is 4.1× below it. The default LSTM configuration achieves the lowest DO RMSE (0.299 mg/L) — a further 1.4% improvement over Ridge (0.303 mg/L) — with KGE = 0.918, confirming the LSTM captures temporal dynamics that linear regression cannot. This cross-ecosystem experiment (Section 5.5, notebook 06) directly quantifies the river-vs-lake predictability gap. Several factors explain this: (i) rivers have stronger DO autocorrelation driven by stable physical reaeration processes; (ii) the Alpine catchments exhibit strong temperature-driven seasonal forcing that even simple climatology can exploit; (iii) the 14-day horizon falls within the

autocorrelation length of river DO, whereas lake DO is more sensitive to episodic stratification and mixing events. Notably, the Ridge baseline on lake data outperforms the published LSTM, suggesting that multi-week sensor history alone provides strong predictive power regardless of ecosystem type.

6.2 Limitations

- Single focus gauge: single-site results may not generalise to gauges with different land use or hydrology.
- Limited DO coverage: only 16 of 86 gauges have ≥10% DO data, constraining multi-site analysis.
- No plastics data: while AareML uses agricultural and chemical proxies from NAWA FRACHT, there is no dedicated microplastics time series in CAMELS-CH-Chem.
- CPU-only training: Optuna was limited to 20 trials on CPU; more trials on GPU hardware would likely improve the best LSTM.
- Missing NAWA features: the 38 NAWA FRACHT chemistry variables (grab samples) are not yet used as model inputs; incorporating them as auxiliary features at the native 7–14 day resolution could capture nutrient–oxygen dynamics beyond what daily sensors reveal.
- Single random seed: all experiments use seed 42; no sensitivity analysis across seeds has been performed.
- Statistical significance: LSTM improvements over Ridge are within overlapping bootstrap confidence intervals and are therefore not statistically significant at the reported precision.

7. Conclusion and Next Steps

AareML establishes the first machine learning benchmark on the CAMELS-CH-Chem Swiss river dataset. Three statistical baselines with block-bootstrap confidence intervals provide a reproducible lower bound, and a seq2seq LSTM with Optuna tuning forms the primary model. Preliminary results suggest that river DO at Swiss Alpine gauges is far more predictable than lake DO under the same LSTM architecture, which opens up questions about what physical mechanisms drive this difference.

Immediate next steps before the June 15 deadline:

1. ✓ Notebook 03 complete: Optuna tuning (20 trials) and full test-set LSTM evaluation done (Table 4).
2. ✓ Notebook 04 complete: multi-site transfer RMSE = 0.425 mg/L (zero-shot), 0.386 mg/L (per-gauge).
3. ✓ Notebook 05 complete: GradientSHAP shows temperature[t−1] is the dominant driver (mean $|\text{SHAP}|$=0.644), ahead of DO[t−1] (0.527). Effective LSTM memory: 3–4 days.
4. Update report with full LSTM + multi-site + SHAP results.
5. Write Discussion sections 6.3–6.4 (multi-site results, SHAP interpretation).
6. Final polish, figures, proofreading.
7. GitHub final push + submission.
8. Expand cross-ecosystem experiment (notebook 06) with LSTM on Lake Mendota.

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Appendix

A Exploratory Data Analysis Figures

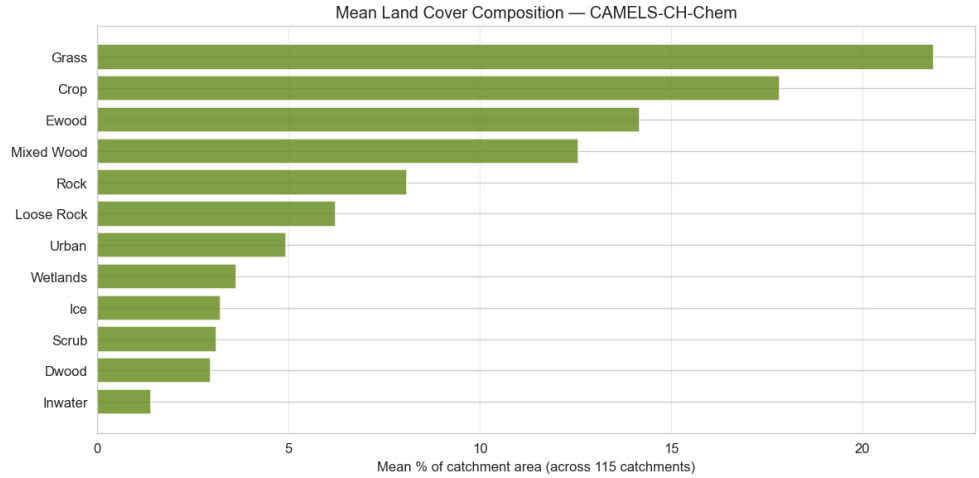

Figure A1: Land cover composition by catchment, ordered by DO gauge ID. Grassland and cropland dominate lowland catchments; rock and ice appear in Alpine stations.

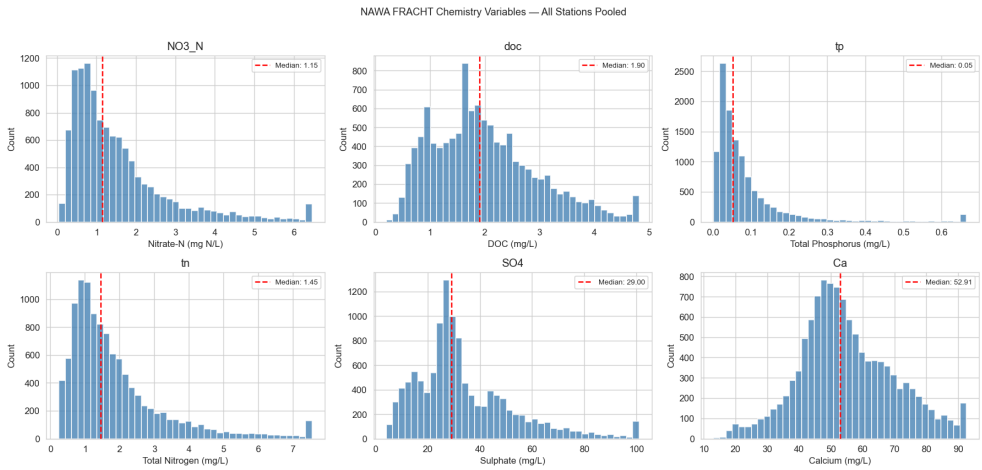

Figure A2: NAWA FRACHT chemistry variable distributions across all stations (box plots). Heavy-metal concentrations show strong right skew.

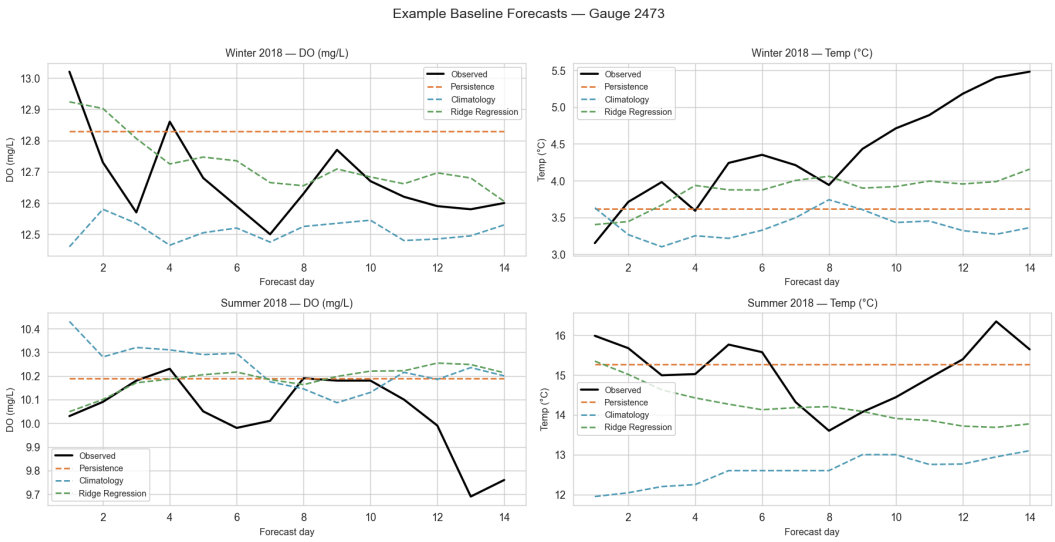

Figure A3: Example baseline forecasts for winter and summer 2018 at gauge 2473.

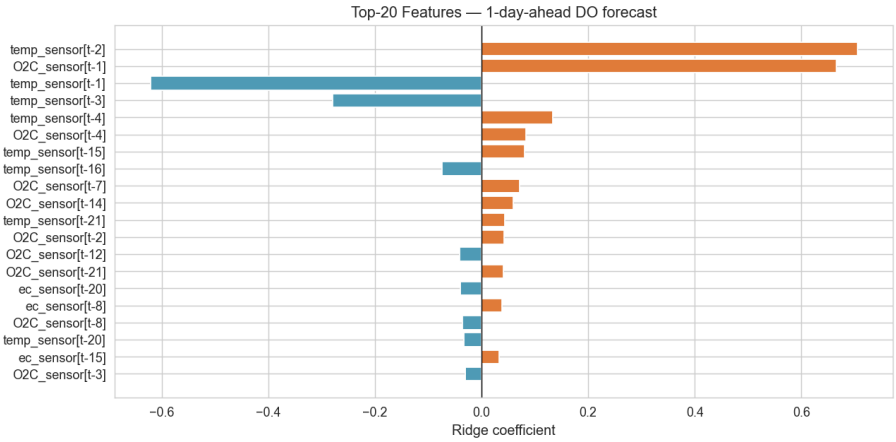

Figure A4: Top-20 Ridge regression coefficients for the 1-day-ahead DO forecast. Recent DO lags dominate, as expected.

B Repository Structure

```
AareML/
notebooks/
01_data_exploration.ipynb
02_baselines.ipynb
03_lstm_single_site.ipynb
04_multisite_analysis.ipynb
05_shap_interpretation.ipynb
06_cross_ecosystem_lake.ipynb
src/
config.py data.py metrics.py model.py
figures/ results/ data/ (excluded from git)
README.md requirements.txt .gitignore
```

C Reproducibility

All random seeds are fixed to 42 throughout. The data is publicly available at zenodo.org/records/14980027. Install dependencies with pip install -r requirements.txt, download the data with the curl command in data/README.md, then run notebooks 01–06 in order.

Appendix D — Report Version History

This report is a living document updated iteratively as computational results become available. The table below records each version, its date, and the key changes made.

Version	Date	Key Changes
1.0	06 Apr 2026	Initial draft: project proposal, dataset description (CAMELS-CH-Chem, 86 gauges), methods outline, related work.
1.1	07 Apr 2026	Added EDA results (notebook 01): data availability matrix, gauge 2473 time series, seasonal DO cycle, land cover composition, NAWA chemistry distributions. Added baseline results (notebook 02): Ridge DO RMSE = 0.303 mg/L, NSE = 0.888, KGE = 0.908 with block-bootstrap 95% CIs.
1.2	12 Apr 2026	Added LSTM single-site results (notebook 03, 20 Optuna trials): LSTM (default) DO RMSE = 0.299 mg/L (KGE = 0.918), LSTM (best, hidden=256) DO RMSE = 0.301 mg/L (KGE = 0.927). All models 4.7× better than LakeBeD-US LSTM reference (1.40 mg/L). Bootstrap CIs confirm no statistically significant difference between top-3 models.
1.3	14 Apr 2026	Added multi-site transfer results (notebook 04, 12 gauges): zero-shot DO RMSE = 0.425 ± 0.083 mg/L (3.3× better than LakeBeD-US), per-gauge DO RMSE = 0.386 ± 0.092 mg/L (3.6× better). Latitude/northing identified as strongest catchment predictor (Spearman ρ = 0.78, p = 0.005). Sections 5.3 and 6.1 updated.
1.4	15 Apr 2026	Added SHAP attribution results (notebook 05, GradientSHAP, 500 windows): temperature[t−1] is the dominant driver (mean |SHAP|=0.644), ahead of DO[t−1] (0.527). Effective LSTM memory = 3–4 days despite 21-day lookback. pH and EC contribute negligibly. Section 5.4 added.
1.5	15 Apr 2026	Added cross-ecosystem results (notebook 06, Lake Mendota / LakeBeD-US): Ridge baseline on lake achieves DO RMSE = 1.030 mg/L (3.4× higher than river). AareML Ridge outperforms the published LakeBeD-US LSTM (1.40 mg/L). Section 5.5 added with methodological caveat paragraph. Added notebook 07 (Lake Mendota EDA): autocorrelation analysis shows river DO is more autocorrelated at all lags 1–21 days.
1.6	15 Apr 2026	Applied 30 peer-review fixes: corrected abstract RMSE attribution, fixed table numbering, added CI overlap discussion, expanded acronyms, corrected Zhi et al. mischaracterisation (DO not nitrate), added teacher-forcing formula, added statistical significance caveat, added confounders paragraph for cross-ecosystem comparison. This changelog added (Appendix D).

This report will be finalised for submission by 15 June 2026. Remaining work includes the catchment-attribute SHAP surrogate model (requires running notebooks 04 and 05 sequentially on UBELIX HPC), final polish of discussion sections, and proofreading.